pfi

Python for Fixed Income

COURSE 5B · TIME SERIES MODELS

Stationarity and Differencing

WEEK 1 · DAY 3

Created by **Jeff Lenamon**

This course is educational content created in a personal capacity. Nothing here is investment advice or a recommendation to buy or sell any security. All portfolio data is synthetic.

Today: does the series behave the same way in every year

Scope

Development years only: 2 Jan 2015 to 30 Dec 2022, 2,001 dates. A test fits nothing for scoring, so it needs no folds. It describes how a series behaved on those dates, not any later date.

Four things

1. What stationary means, in five simulated series
2. ADF and KPSS, every setting, read together
3. Why one wandering series on another lies
4. Difference until it settles

Day 2's clue: the training mean was a dozen times worse, because a level wanders.

Names for a test result

A test result is a p-value, a reading, a verdict on the series. A model's output, when we get to models:

Allowed names

fitted change in bp

the model's one-step value

the last value carried forward

Never called

a forecast or a prediction

a projection or an outlook

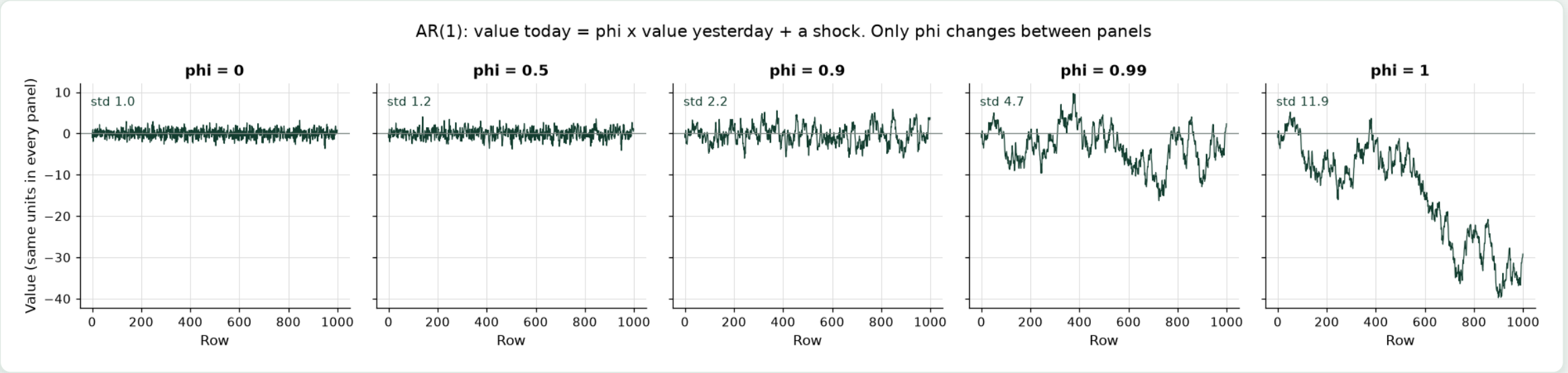
a signal, a call, a view on rates

Nothing today says what a yield will do.

How stationarity works

The problem	A model fitted on one stretch must carry to another, so the series must behave the same way throughout
Stationary	The same average, spread, and link to its past in every year: it keeps coming back to its average
A unit root	Each value is the last plus a shock never undone: it wanders, and its past average describes nothing
AR(1)	Value today = phi x value yesterday + a shock: below 1 is stationary, at 1 a random walk
You control	phi and the size of the shocks; what comes out is a series whose average and spread you can measure

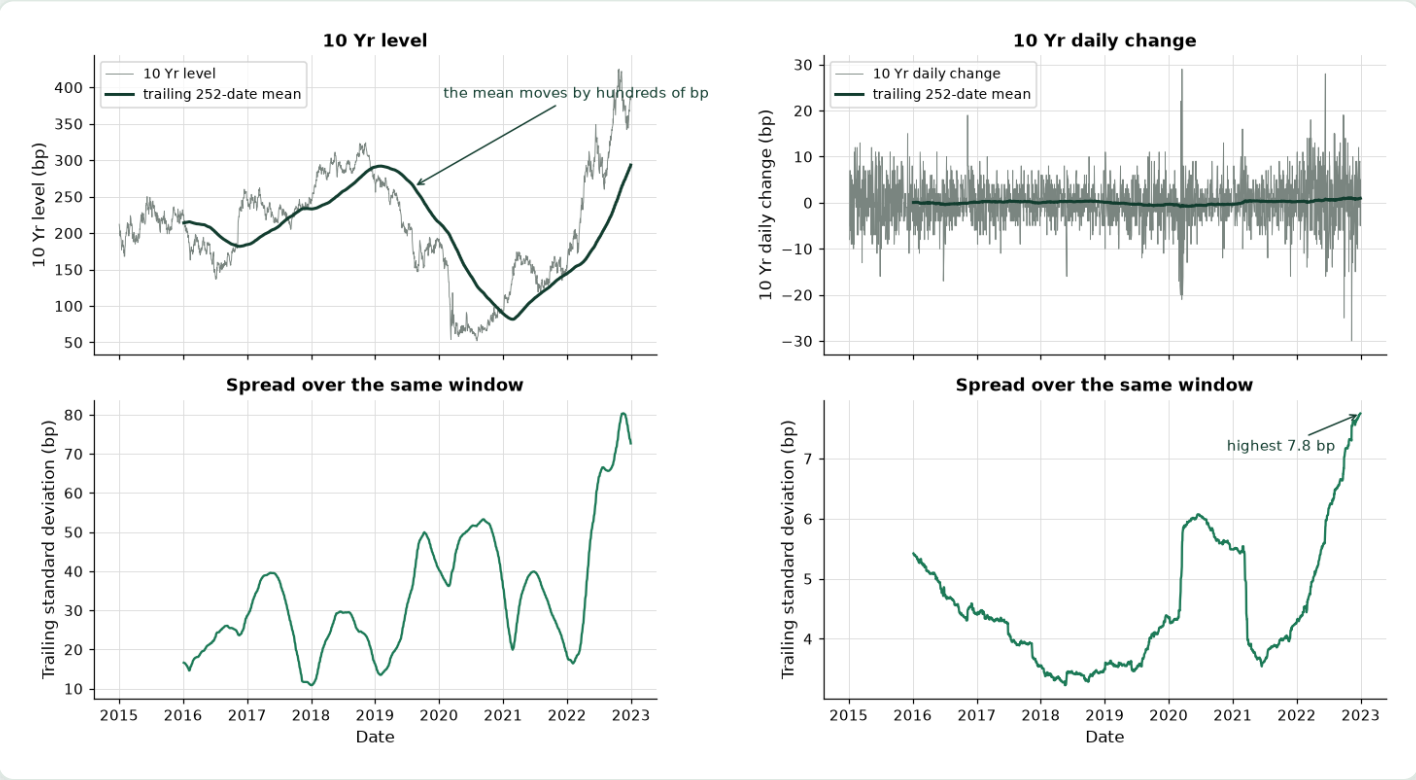
Five values of phi, one set of shocks



Same shocks in every panel, so any difference is phi. At phi = 0.99 the series is stationary on paper and behaves like a walk over 1,000 rows.

That is why a test is needed, and why a test can be unsure near phi = 1.

A level and a change, with a trailing window



What it shows

The level's trailing mean moves 212 bp. The change's moves 1.91 bp, around zero. The change's spread ran from 3.23 bp (21 May 2018) to 7.75 bp (30 Dec 2022).

A trailing window includes its own date: for scoring that date it needs shift(1) first.

How the ADF test works

The problem	Does yesterday's level pull today's change back toward a level?
The regression	The day's change on yesterday's level, a constant, and some lagged changes
The statistic	The coefficient on yesterday's level over its standard error, against critical values more negative than a normal table's
Null hypothesis	A unit root: a small p-value (below 0.05) rejects it and says stationary
You control	The regression (c, ct, n) and the lags (maxlag, autolag)

Every setting of adfuller

Setting	Default	Tune first
regression	'c'	yes
maxlag	None	yes
autolag	'AIC'	yes
store	False	no
regresults	False	no
result_object	None	no

The two that matter

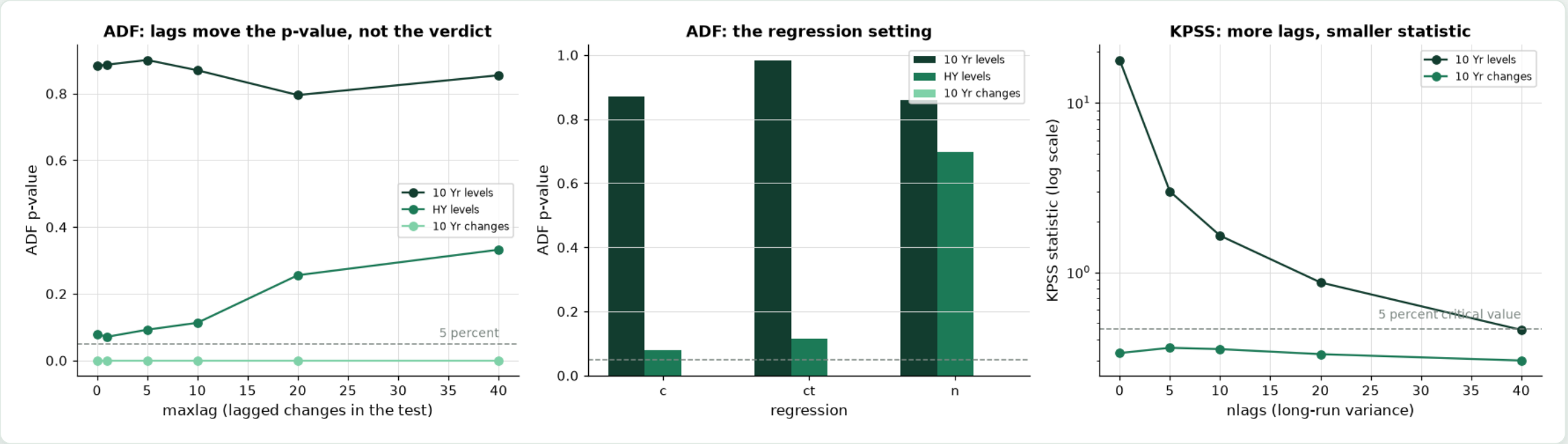
regression: c a constant, ct a constant and a trend, n neither. Lags: maxlag, with autolag choosing up to it (AIC by default). Defaults read from the installed statsmodels. Always pass result_object=True: without it statsmodels warns.

A warning is a defect in this course: fix it at its source.

How the KPSS test works

The problem	The same question turned round: its null hypothesis is that the series is stationary
The running total	Take the average (or trend) away, add up what is left, and compare its size with a stationary series'
Reading it	A wandering series' deviations pile up, the statistic is large, the null is rejected
You control	The regression (c, ct) and nlags, the lags in the long-run variance
What comes out	A statistic, a p-value from a table that runs only from 0.01 to 0.1, and critical values

Moving the settings



Left: the 10 Yr level stays far above the 5 percent line at every maxlag, its change sits at zero. Middle: a trend does not rescue the levels. Right: more KPSS lags can hide a wandering series.

Choose the settings from the series and say so; never from the one that gives the answer you wanted.

Two tests, four readings

Case	ADF	KPSS	Reading
1	rejects	does not reject	stationary
2	does not reject	rejects	a unit root
3	rejects	rejects	shifts in mean or spread
4	does not reject	does not reject	not enough evidence

phi	ADF p	KPSS p	Rejects
0	0.0000	True	False
0.5	0.0000	True	False
0.9	0.0000	True	False
0.99	0.0053	True	True
1	0.5787	False	True

On series whose truth we know: at phi = 0.99 the tests disagree.

When they disagree, look at the series and the settings, and say what you decided.

Five series, levels and changes

Series	Form	ADF p	KPSS p	Reading
2 Yr	levels	0.9779	0.0100	unit root
2 Yr	changes	0.0000	0.0100	both reject
10 Yr	levels	0.8699	0.0194	unit root
10 Yr	changes	0.0000	0.1000	stationary
2s10s	levels	0.7879	0.0100	unit root
2s10s	changes	0.0000	0.1000	stationary
IG	levels	0.5529	0.0100	unit root
IG	changes	0.0000	0.1000	stationary
HY	levels	0.0793	0.0100	unit root
HY	changes	0.0000	0.1000	stationary

Reading it

All five levels read as a unit root. Four of five changes read as stationary. The 2 Yr's changes: both tests reject, so look for a shift in mean or variance.

KPSS p of 0.01 means 0.01 or smaller; 0.1 means 0.1 or larger.

Why the 2 Yr's changes fail KPSS

Year	Mean change (bp)	Std dev (bp)
2015	0.16	3.40
2016	0.06	3.35
2017	0.28	2.54
2018	0.24	2.96
2019	-0.36	4.47
2020	-0.58	3.33
2021	0.24	2.35
2022	1.48	8.00

One year is unlike the others

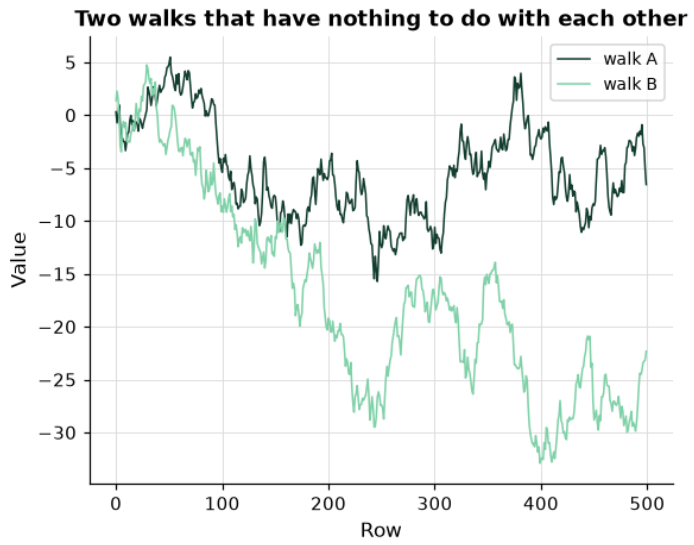
2022: mean 1.48 bp a day, standard deviation 8.00 bp. A test that assumes one average and one spread over eight years sees that.

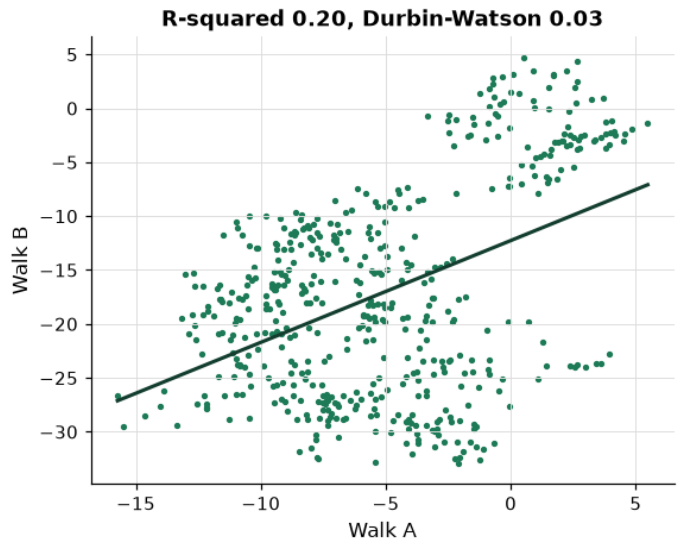
A unit root removed is not a series that behaves the same in every year.

How a spurious regression works

The problem	A regression says how well a line explains one series from another
Stationary pair	R-squared near zero, as it should be
Wandering pair	Each drifts, a stretch where both drift together looks like a link: R-squared high, residuals one long wander
The tell	A Durbin-Watson statistic near 0
The fix	Regress changes on changes, or test for a link that survives differencing (day 9)

Two walks with nothing to do with each other







One pair: R-squared 0.200, Durbin-Watson 0.025. Across 1,000 pairs the median R-squared of independent walks is 0.182, and 37.7% pass 0.3; white noise has a median of 0.0010.

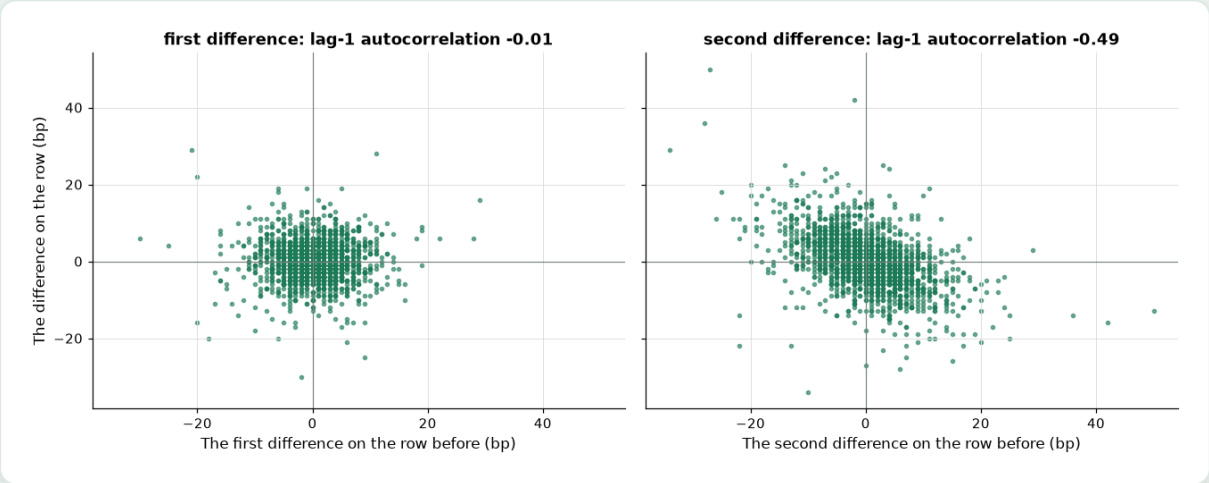
On real data: the 10 Yr's level explains 13.9% of the invented IG spread's level, and its changes 0.01% of its changes.

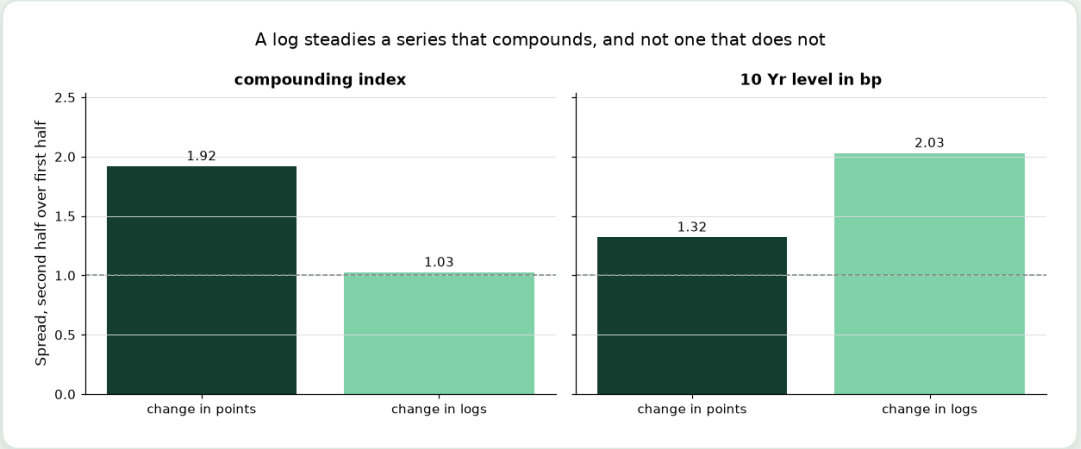
A high R-squared between two levels is not evidence of a relationship.

How differencing works

The problem	A unit-root series is a running total of stationary shocks
A first difference	Each value less the one before it undoes the running total and leaves the shocks
Over-differencing	A second difference of a stationary series adds noise and a lag-1 correlation near -0.5
A log	Helps when moves scale with the level (a compounding index); a change in logs is a percentage move
What comes out	A series one value shorter per difference, in bp or in log points

Over-differencing, and when a log helps





Lag-1 autocorrelation -0.007 for the first difference and -0.485 for the second; its standard deviation is 7.14 bp against 5.03. The second difference passes ADF too: the test cannot say you differenced too much.

Compounding index: 1.92 in points, 1.03 in logs. The 10 Yr: 1.32 in points, 2.03 in logs.

Difference once, test, and stop. Yields and spreads in bp are differenced, not logged.

The stationarity checklist

Step	Question	Tool
1	Does the level wander?	a trailing mean and standard deviation, drawn
2	Does the spread of the changes move?	the same, on the changes
3	What do ADF and KPSS say?	adf_summary and kpss_summary, side by side
4	Did differencing once settle it?	both tests again, on the changes
5	Did differencing twice add noise?	the lag-1 autocorrelation of the second difference

Run it before every model: days 4 and 5 start from its answer.

Excel side by side

Excel side by side (tested in Excel)

`=AVERAGE(B1751:B2002)`	last 252 levels, mean 293.37
`=STDEV.S(B1751:B2002)`	spread 72.65
`=SLOPE(B3:B2002,B2:B2001)`	AR(1) phi 0.9992
`=SLOPE(D3:D2002,B2:B2001)`	Dickey-Fuller slope -0.000832
`=INDEX(LINEST(D3:D2002,B2:B2001,TRUE,TRUE),1,1)`	slope (gamma)
`=INDEX(LINEST(D3:D2002,B2:B2001,TRUE,TRUE),2,1)`	std error: ratio -0.5456
`=RSQ(B2:B2002,C2:C2002)`	R-squared, levels 0.1389
`=RSQ(D3:D2002,E3:E2002)`	changes 0.000093

Excel has no ADF or KPSS test and no table of the critical values: it has every piece the tests are made of.

Save the day: a branch

1. Init

The work folder, local settings, the guard

2. Commits

The code, then stationarity.csv on main

3. Branch

git switch -q -c trend, and the experiment's file

4. Diff

git diff --stat main trend, then back to main

```
Day 3: ADF with a trend on the levels
Day 3: tskit with adf_summary and kpss_summary
```

A side experiment gets its own branch: the main line stays as it was.

The AI check

The docstring is the prompt. The code that comes back runs, and its number looks fine.

1

Docstring

Which dates fit, which dates score, what may be filled: what you asked for.

2

Code back

Written for the course in the style of an AI answer. Read it.

3

Test first

Assert what the docstring promises. Run it: it fails.

4

Fix

Find the rule it broke, fix it, run the test again.

Today: the p-value is compared the wrong way round: the random walk is called stationary (True) and the noise is not (False).

Code that runs is not the same as code that is right.

What you can do now

- What stationary means, and what phi does to a series
- ADF and KPSS: each null, each setting, read together
- A spurious regression: a high R-squared between levels proves nothing
- Difference once, test, and stop; logs only for compounding series
- The stationarity checklist, before every model

Week 1, Day 4 Autocorrelation: the ACF and the PACF